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Chapter 3 — Cells and Subcellular Organelles

Complete Chapter Notes · Textbook topics converted into board-style long questions, with important conceptual short questions inside every topic · PECTAA 2025–26, Punjab English Medium

This chapter follows the textbook topics 3.1 to 3.8. Each topic is converted into board-style long questions, and each topic ends with conceptual short questions. Direct definitions are omitted.

Board pattern (Punjab 2026, new syllabus): Chapter 3 carries 1 MCQ, 3 short questions in Q.2, and part (a) of a long question in Q.6. Shorts are 2 marks; longs are 8 marks.

TOPIC 3.1 — CELLS — THE BASIC UNIT OF LIFE

Textbook topic 3.1 · printed pages 48–49

Q.1 Explain why the cell is considered the basic unit of life.

Board long question · 8 marks

Cells are the basic unit of life, making up every living organism. In unicellular organisms like amoebas and bacteria, a single cell carries out all the functions necessary for life. Multicellular organisms, such as plants and animals, are composed of numerous specialized cells that work together to sustain life.

All living organisms show the seven basic properties of life: **movement, respiration, homeostasis, growth, reproduction, excretion and nutrition**. Cells perform all these fundamental activities.

- **Movement:** Cells can move, e.g., sperm cells move with flagella and white blood cells travel through the bloodstream. Inside cells, organelles move to carry out vital functions.
- **Nutrition:** Cells obtain nutrients from the environment to produce energy, build cellular structures and drive biochemical reactions.
- **Respiration:** Cells generate energy by breaking down glucose to release ATP, the energy currency that powers cellular activities.
- **Excretion:** Cells remove waste products through diffusion and active transport, preventing toxic buildup.
- **Homeostasis:** Cells maintain a stable internal environment by regulating movement of substances across their membranes.
- **Growth:** Cells grow by taking in nutrients and converting them into cellular components.
- **Reproduction:** Cells reproduce through mitosis and meiosis. Mitosis produces identical daughter cells for growth and repair; meiosis creates gametes for sexual reproduction.

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Important Conceptual Short Questions — Topic 3.1

S1. Why is a unicellular organism able to survive with only one cell?

A single cell carries out all the functions necessary for life, including nutrition, respiration, excretion, homeostasis, growth, reproduction and movement.

S2. How do specialized cells in multicellular organisms contribute to life?

They perform specific functions and work together to sustain the organism, allowing division of labour among tissues and organs.

S3. Why is ATP called the energy currency of the cell?

It is produced during respiration by breaking down glucose and powers the cell's metabolic activities.

TOPIC 3.2 — CELL THEORY

Textbook topic 3.2 · printed pages 49–51

Q.2 Describe the development of cell theory, its main points, validation and exceptions.

Board long question · 8 marks

Development

- **Robert Hooke (1665):** Examined a thin slice of cork and observed "many little boxes", which he named **cellulae**.
- **Anton van Leeuwenhoek (1673):** Made a better microscope and observed living cells in pond water, calling them **animalcules**.
- **Robert Brown (1831):** Discovered the nucleus in the cell.
- **Matthias Schleiden (1838):** Concluded that all plants are composed of cells.
- **Theodor Schwann (1839):** Concluded the same for animals.
- **Rudolf Virchow (1885):** Observed that all cells come from other cells; Louis Pasteur (1862) provided experimental proof.

Three essential points

1. All living organisms are composed of one or more cells.
2. Cells are the basic units of structure and function in an organism.
3. Cells come only from the division of pre-existing cells.

Validation

Light and electron microscopy visualize cell structures; live-cell imaging and genetic studies track cell replication; DNA sequencing shows common genetic material and metabolic pathways; cell culture and tissue engineering demonstrate cellular growth, differentiation and reproduction.

Exceptions

- **Viruses, prions and viroids** are not made of cells and cannot carry out life processes independently.

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- **Mitochondria and chloroplasts** have their own DNA and can replicate independently of the nucleus.
- Some fungi and algae have **multiple nuclei within a shared cytoplasmic mass**.
- Vertebrate **muscle cells** can fuse to form multinucleated fibres.

Important Conceptual Short Questions — Topic 3.2

S1. Why do viruses challenge cell theory?

They are not made of cells and cannot carry out life processes independently. They require a host cell to replicate and are considered at the border of living and non-living entities.

S2. How do mitochondria and chloroplasts challenge the idea that all cellular activity is controlled by the nucleus?

They have their own DNA and can replicate independently of the cell's nucleus, suggesting they may have originated from free-living prokaryotic cells.

S3. Why do multinucleated muscle fibres challenge traditional cell theory?

They form by fusion of muscle cells and contain many nuclei in a shared cytoplasm, blurring the concept of a single cell as the basic unit.

S4. How did Pasteur support Virchow's idea?

Pasteur provided experimental proof that cells come only from pre-existing cells, supporting Virchow's observation.

TOPIC 3.3 — MICROSCOPY

Textbook topic 3.3 · printed pages 51–54

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Q.3 Differentiate between light and electron microscopy. Explain magnification and resolution.

Board long question · 5 + 3 marks

Feature	Light microscope	Electron microscope
Source	Light	Electron beam (approx. 1 nm)
Magnification	Usually 500X to 1500X	Direct 16000X; photographic up to 1000000X
Resolution	About 200 nm	May be 0.2 nm
Specimen	Both live and dead specimens can be seen	Only dead and dried specimens
Preparation	Takes about a few hours	Takes several days
Image	Seen through ocular lens	Received on a zinc sulphate fluorescent screen

Magnification is the ability to make an object appear larger. Total magnification is determined by multiplying the magnification of the ocular lens by that of the objective lens. **Resolution** is the ability to distinguish two close points as separate. A light microscope can resolve details down to approximately 200 nm, while an electron microscope has much higher resolution.

Important Conceptual Short Questions — Topic 3.3

S1. Why can an electron microscope reveal more detail than a light microscope?

It uses an electron beam with a much shorter wavelength than light, giving higher resolution (about 0.2 nm) and much higher magnification.

S2. Why can living specimens be observed under a light microscope but not an electron microscope?

Electron microscopy requires dead, dried specimens and extensive preparation, while light microscopy can observe both live and dead specimens.

S3. Differentiate between magnification and resolution.

Magnification makes an object appear larger. Resolution is the ability to distinguish two close points as separate; it determines the clarity and detail of the image.

TOPIC 3.4 — STRUCTURE OF CELL

Textbook topic 3.4 · printed pages 54–78

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Q.4 Describe the structure and functions of the cell wall.

Board long question · 8 marks

The cell wall is a more or less solid layer surrounding a cell. It is found in bacteria, fungi, plants and algae. When a cell wall is removed using cell wall degrading enzymes, the remaining components of the cell are called a **protoplast**.

- **Primary wall:** Composed of polysaccharides — cellulose, hemicellulose and pectin. Cellulose microfibrils are aligned at all angles and held together by hydrogen bonds.
- **Middle lamella:** A gelatinous layer that separates and holds the primary walls of neighbouring cells. It contains magnesium and calcium pectates.
- **Secondary wall:** Formed in some plant cells after maturation, between the protoplast and primary wall. It contains lignin, cellulose and hemicellulose, making it more rigid.
- **Plasmodesmata:** Small channels that directly connect the cytoplasm of neighbouring plant cells, allowing certain molecules to pass and helping cellular communication.

Functions: The cell wall provides rigidity for structural and mechanical support, maintains cell shape and direction of growth, prevents expansion when water enters the cell, protects against pathogens and the environment, and stores carbohydrates.

Composition in different groups: Algal walls contain cellulose and glycoproteins; fungal walls contain chitin; prokaryotic walls contain peptidoglycan; archaebacterial walls contain different polysaccharides and proteins with no peptidoglycan.

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Q.5 Explain the fluid mosaic model of the plasma membrane.

Board long question · 8 marks

The fluid mosaic model was proposed by **S.J. Singer and Garth Nicolson in 1972**. According to this model, the basic foundation of the plasma membrane is a **lipid bilayer** made of phospholipids, in which a collection of proteins float.

- Phospholipids have a **hydrophilic phosphate head** and **hydrophobic fatty acid tails**. In the bilayer, the hydrophobic tails face inward and the hydrophilic heads face outward toward the cytoplasm or extracellular environment.
- The interior of the lipid bilayer is hydrophobic and repels water-soluble molecules. Proteins provide passageways across the membrane.
- In eukaryotes, **cholesterol** molecules are wedged into the phospholipid bilayer and keep membrane fluidity at low temperatures.
- Some proteins float within the bilayer (**integral proteins**), while others adhere to the surfaces (**peripheral proteins**).

Functions of membrane proteins: selective transport as channels or active transport molecules; attachment to the cytoskeleton and external fibres;

identification marks by attaching with sugars on the exterior surface; and receptors that bind messenger molecules.

Q.6 Describe the structure and functions of the nucleus.

Board long question · 8 marks

- **Nuclear envelope:** A double membrane enclosing the nucleus. The outer nuclear membrane is continuous with the endoplasmic reticulum.
- **Nuclear pores:** Tiny openings in the nuclear envelope that allow movement of molecules between nucleus and cytoplasm. They are permeable to small molecules and regulate transport of RNA and proteins.
- **Nucleolus:** A dense region where ribosomal RNA is synthesized and ribosomal subunits are assembled.
- **Chromatin:** DNA associated with proteins. During cell division it condenses to form chromosomes.

The nucleus controls cellular activities by storing genetic information and regulating gene expression. Human diploid cells have **46 chromosomes**. Chromosome number is species-specific and does not correlate with complexity of the organism.

Q.7 Describe the structure and functions of endoplasmic reticulum and ribosomes.

Board long question · 5 + 3 marks

Endoplasmic Reticulum

The ER is a network of flattened sacs and branching tubules called **cisternae**. It is connected to the nuclear envelope and provides a pipeline between nucleus and cytoplasm. It manufactures, processes and transports biochemical compounds.

Type	Structure	Function
Rough ER	Surface covered with ribosomes	Production and processing of proteins; adds sugars to proteins; sends them to Golgi apparatus
Smooth ER	Surface lacks ribosomes	Production of lipids, carbohydrate metabolism, detoxification of drugs and poisons; stores calcium and releases it to trigger muscle contraction

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Ribosomes

Ribosomes are tiny granular structures composed of about **60% rRNA and 40% protein**. They are not membrane-bound. In eukaryotes they are attached to RER or free in cytoplasm; in prokaryotes they are free in cytoplasm. Eukaryotic ribosomes are **80S** (60S + 40S), while prokaryotic ribosomes are **70S** (50S + 30S). They serve as the protein production machinery of the cell.

Q.8 Describe the structure and functions of mitochondria. Why are they called the power generators of the cell?

Board long question · 8 marks

Mitochondria are rod-shaped organelles bounded by **two membranes**. There is a narrow **intermembrane space** between them. Beneath the inner membrane is a larger internal **matrix**. The outer membrane is smooth and acts like a sieve. The inner membrane is highly convoluted and forms infoldings called **cristae**, which increase surface area.

The inner surface of the cristae has knob-like extensions called **F1 particles**, which are actually the enzyme **ATP synthase**. Other complexes in the inner membrane serve as electron carriers in the electron transport chain.

Mitochondria have their own **circular DNA**, all kinds of RNA and **70S ribosomes**, and can replicate independently of the cell. They are the sites of **cellular respiration** and generate **ATP** from oxygen and nutrients. Enzymes in the matrix catalyse steps such as the Krebs cycle, while electron transport chain proteins are found on the inner membrane.

Q.9 Describe the structure and functions of chloroplasts.

Board long question · 8 marks

Chloroplasts are ellipsoid-shaped organelles enclosed by **two membranes**. The area between the membranes is the intermembrane space. Inside the inner membrane is a semi-fluid called **stroma**, which contains dissolved enzymes.

The inner membranes form disk-shaped sacs called **thylakoids**. Many thylakoids form stacks called **grana** (singular: granum). The non-green compartments connecting two grana are called **lamellae** or stroma lamellae. Chloroplasts contain chlorophyll **a** and **b**, which absorb light energy for photosynthesis.

Light is absorbed by chlorophyll molecules embedded in the thylakoid disks. When they absorb light, they emit electrons and ATP is formed. Using this ATP, low-energy carbon dioxide is transformed into high-energy compounds like glucose in the stroma. Like mitochondria, chloroplasts have their own DNA and reproduce independently of the cell.

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Q.10 Describe the structure and functions of the Golgi apparatus.

Board long question · 8 marks

The Golgi apparatus consists of five to eight cup-shaped, membrane-covered sacs called **cisternae** stacked over each other. Each Golgi stack has two distinct faces: the **cis face** near the endoplasmic reticulum and the **trans face** near the plasma membrane.

The Golgi apparatus is the **distribution and shipping department** of the cell. It modifies proteins and lipids built in the ER and prepares them for export outside the cell or for transport to other locations. Vesicles from the ER fuse with the cis face and release their molecules into the Golgi. Enzymes in the Golgi lumen convert them into glycoproteins and glycolipids. The products are extruded from the trans face in vesicles and directed to their final destination. Some products become lysosomes. In plant cells, the Golgi apparatus also produces pectin and other polysaccharides needed for plant structure and metabolism.

Q.11 Describe lysosomes, peroxisomes and glyoxysomes.

Board long question · 8 marks

Lysosomes

Lysosomes are spherical organelles bounded by a single membrane. They serve as **digestive compartment** of the cell and contain about **40 different hydrolytic enzymes**. These enzymes require an acidic environment (pH about 4.8). The lysosomal membrane protects the rest of the cell from these enzymes. Lysosomes break down waste products, worn-out organelles (**autophagy**), and foreign material. They also function in programmed cell death, e.g., destruction of tadpole tail cells during metamorphosis.

Peroxisomes

Peroxisomes are single membrane-bound organelles containing oxidative enzymes. They detoxify harmful substances, e.g., alcohol in liver cells. Some contain **catalase**, which breaks down hydrogen peroxide into water and oxygen.

Glyoxysomes

Glyoxysomes are similar to peroxisomes but are found only in plant cells. They contain enzymes that convert **lipids into carbohydrates**. They are abundant in lipid-rich seeds and provide energy for seed germination.

Q.12 Describe the structure and functions of vacuoles, cytoskeleton, centrioles, cilia and flagella.

Board long question · 8 marks

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Vacuoles

Vacuoles are membrane-bounded sacs. In mature plant cells, a single large central vacuole provides structural support, storage, waste disposal, protection and growth. It is enclosed by the **tonoplast** and contains **cell sap**. The central vacuole controls **turgor pressure**, which maintains rigidity of the plant cell. Animal cell vacuoles are smaller and used for temporary storage or transport.

Cytoskeleton

- **Microfilaments:** Solid rods made of actin; help cells change shape and move, and enable a dividing cell to pinch into two.
- **Microtubules:** Hollow cylinders made of alpha- and beta-tubulin; give shape, serve as highways for organelle transport, form cilia/flagella and spindle fibres.
- **Intermediate filaments:** Made of proteins such as vimentin or keratin; maintain cell shape and rigidity and anchor organelles.

Centrioles, Cilia and Flagella

Centrioles are hollow cylindrical organelles made of **nine triplets of microtubules**. They occur in pairs at right angles near the nuclear envelope and are associated with assembly of microtubules, including spindle fibres. In ciliated/flagellated cells, they form the **basal body**. Plants and fungi lack centrioles and basal bodies.

Cilia and flagella are locomotor appendages. Cilia are short and numerous; flagella are longer and fewer. Their core, the **axoneme**, has a **9+2 arrangement**: two central microtubules surrounded by nine doublet microtubules. Dynein molecules bridge adjacent doublets. The basal body has nine triplets of microtubules.

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Important Conceptual Short Questions — Topic 3.4

S1. Why does the secondary cell wall provide more rigidity than the primary wall?

The secondary wall contains lignin in addition to cellulose and hemicellulose. Lignin makes it more rigid than the primary wall.

S2. How do plasmodesmata help in cellular communication?

They are small channels connecting the cytoplasm of neighbouring plant cells, allowing certain molecules to pass directly from one cell to another.

S3. Why is the plasma membrane described as fluid and mosaic?

It is fluid because phospholipids and proteins can move within the bilayer, and mosaic because proteins are embedded in a patchwork pattern within the lipid bilayer.

S4. How does cholesterol affect the plasma membrane?

Cholesterol molecules wedged into the phospholipid bilayer keep membrane fluidity at low temperatures.

S5. Why are membrane proteins necessary for transport of water-soluble molecules?

The hydrophobic interior of the lipid bilayer repels water-soluble molecules. Membrane proteins provide channels or carriers for their passage.

S6. How does the structure of mitochondria support ATP production?

The inner membrane forms cristae that increase surface area for electron transport chain complexes and F1 particles (ATP synthase), while matrix enzymes catalyse the Krebs cycle.

S7. Why are mitochondria and chloroplasts considered semi-autonomous organelles?

They have their own circular DNA, RNA and 70S ribosomes and can replicate independently of the cell, suggesting an endosymbiotic origin.

S8. How do thylakoids and grana increase the efficiency of photosynthesis?

Thylakoid membranes contain chlorophyll and provide a large surface area for light absorption. Stacking into grana further increases the area available for light reactions.

S9. Differentiate between the cis and trans faces of the Golgi apparatus.

The cis face is near the endoplasmic reticulum and receives vesicles from it. The trans face is near the plasma membrane and ships modified products to their final destination.

S10. Why are lysosomes called suicide bags of the cell?

They contain powerful hydrolytic enzymes that can digest cellular components. If released, they can break down the cell itself, and they also participate in programmed cell death.

S11. How do lysosomes protect the rest of the cell from their digestive enzymes?

They are bounded by a single membrane that keeps the enzymes enclosed. The enzymes require an acidic pH; if the lysosome ruptures, the enzymes become inactive in the neutral cytoplasm.

S12. Why are glyoxysomes important during seed germination?

They convert stored lipids into carbohydrates, providing energy for the germinating seed.

S13. How does the central vacuole maintain plant rigidity?

It stores cell sap and controls turgor pressure. When it swells with water, it pushes against the cell wall, maintaining structural integrity.

S14. Differentiate between microfilaments and microtubules.

Microfilaments are solid rods made of actin and help in cell shape change and movement. Microtubules are hollow cylinders made of tubulin and help in organelle transport, spindle formation and cilia/flagella structure.

S15. Why do plant cells lack centrioles but still form spindle fibres?

Plants and fungi lack centrioles and basal bodies; their microtubules and spindle fibres are organized from other structures in the cytoplasm.

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S16. How does the 9+2 arrangement help cilia and flagella move?

Dynein arms bridge adjacent microtubule doublets and cause them to slide, producing bending movements of cilia and flagella.

TOPIC 3.5 — PROKARYOTIC AND EUKARYOTIC CELLS

Textbook topic 3.5 · printed pages 78–79

Q.13 Differentiate between prokaryotic and eukaryotic cells.

Board long question · 8 marks

Characteristic	Prokaryotic cell	Eukaryotic cell
Distinct nucleus	Absent; nucleoid present	Present; true nucleus with nuclear envelope
Membrane-bound organelles	Absent	Present (mitochondria, ER, Golgi, etc.)
DNA	Single circular DNA without histones	Multiple linear chromosomes with histones
Ribosomes	70S	80S (70S in mitochondria and chloroplasts)
Cell wall	Usually peptidoglycan (bacteria)	Cellulose in plants, chitin in fungi; absent in animals
Cytoskeleton	Absent	Present
Examples	Bacteria and archaea	Protists, fungi, plants and animals

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Important Conceptual Short Questions — Topic 3.5

S1. Why are prokaryotic cells considered simpler than eukaryotic cells?

They lack a true nucleus and membrane-bound organelles. Their DNA is a single circular molecule in the nucleoid, and they have 70S ribosomes.

S2. How does the presence of histones differentiate eukaryotic DNA from prokaryotic DNA?

Eukaryotic DNA is associated with histone proteins and organized into linear chromosomes, while prokaryotic DNA is circular and lacks histones.

S3. Why do mitochondria and chloroplasts contain 70S ribosomes even in eukaryotic cells?

They are believed to have originated from free-living prokaryotes through endosymbiosis, so they retain prokaryote-like ribosomes.

TOPIC 3.6 — CELL SIGNALLING

Textbook topic 3.6 · printed pages 80–82

Q.14 Explain the steps of cell signalling and differentiate between protein/peptide and steroid signalling pathways.

Board long question · 5 + 3 marks

1. **Signal reception:** A signal molecule (ligand) binds to a specific receptor on the target cell membrane or inside the cell.
2. **Signal transduction:** The receptor undergoes a conformational change and activates an intracellular signalling pathway. Second messengers such as cAMP, calcium ions and IP3 amplify and transmit the signal.
3. **Cellular response:** The pathway leads to changes in gene expression, metabolism, enzyme activity, ion channels, cell growth, division, differentiation or apoptosis.

Feature	Protein/peptide signalling	Steroid signalling
Nature of ligand	Water-soluble; cannot pass through plasma membrane	Lipophilic; can diffuse through plasma membrane
Receptor location	On plasma membrane	In cytoplasm or nucleus
Mechanism	Ligand binds membrane receptor; second messengers like cAMP are generated	Hormone binds intracellular receptor; receptor-hormone complex binds DNA
Response	Changes in metabolism, growth, division or apoptosis	Regulates transcription of target genes

Important Conceptual Short Questions — Topic 3.6

S1. Why can steroid hormones pass through the plasma membrane but protein hormones cannot?

Steroid hormones are lipophilic and can diffuse through the lipid bilayer. Protein/peptide hormones are water-soluble and cannot pass through the hydrophobic interior of the membrane.

S2. How do second messengers amplify a signal?

They transmit the signal from the receptor to target molecules inside the cell, creating a signalling cascade that amplifies the original signal.

S3. Differentiate between the receptor locations for protein and steroid hormones.

Protein/peptide hormones bind receptors on the plasma membrane. Steroid hormones bind intracellular receptors in the cytoplasm or nucleus.

S4. How does cell signalling lead to apoptosis?

The signal transduction pathway can turn specific genes on or off, leading to programmed cell death as one of the cellular responses.

TOPIC 3.7 — MEMBRANE TRANSPORT MECHANISMS

Textbook topic 3.7 · printed pages 82–87

Q.15 Describe passive transport mechanisms across the plasma membrane.

Board long question · 8 marks

Diffusion

Diffusion is the net movement of a substance from an area of higher concentration to lower concentration, along the concentration gradient. It does not require energy.

Facilitated Diffusion

Some molecules move down their concentration gradient with the help of transport proteins. It is passive because no energy is used. **Channel proteins** form hydrophilic channels (e.g., ion channels, aquaporins). **Carrier proteins** bind the molecule, undergo a conformational change and move it across the membrane (e.g., glucose transporters).

Osmosis

Osmosis is the diffusion of water across a selectively permeable membrane from a region of higher water concentration to lower water concentration. Water moves from a **hypotonic** solution to a **hypertonic** solution. It is crucial for cell turgor and water balance. **Aquaporins** facilitate rapid water transport.

> **Important**
>
> **Reverse osmosis** applies external pressure to push water through a semi-permeable membrane from higher to lower solute concentration, removing dissolved salts and impurities. It is used in water purification and desalination.

Q.16 Describe active transport and bulk transport across the plasma membrane.

Board long question · 8 marks

Active Transport

Active transport moves substances from lower concentration to higher concentration using energy. Carrier proteins such as the **sodium-potassium pump** use ATP to move Na+ out of the cell and K+ into the cell, maintaining higher K+ and lower Na+ inside resting nerve cells.

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Endocytosis

In endocytosis, bulky material is moved into the cell. The plasma membrane invaginates, encloses the material and forms a vesicle that detaches into the cytoplasm. **Phagocytosis** takes in solid material; **pinocytosis** takes in liquid droplets. In **receptor-mediated endocytosis**, specific receptor proteins pick up material, e.g., liver cells take up LDL cholesterol from blood.

Exocytosis

Exocytosis exports bulky material out of the cell. The material is packed into a vesicle, which moves to the plasma membrane, fuses with it and releases its contents into the extracellular environment.

Important Conceptual Short Questions — Topic 3.7

S1. Why does diffusion not require energy?

It moves substances along their concentration gradient, from higher to lower concentration, without expenditure of cellular energy.

S2. Differentiate between simple diffusion and facilitated diffusion.

Simple diffusion occurs directly through the lipid bilayer. Facilitated diffusion uses channel or carrier proteins but still moves substances down their concentration gradient without energy.

S3. How does osmosis differ from diffusion?

Osmosis is specifically the diffusion of water across a selectively permeable membrane, while diffusion refers to movement of any substance along its concentration gradient.

S4. Why does water move from a hypotonic solution to a hypertonic solution?

A hypotonic solution has lower solute concentration and higher water concentration, while a hypertonic solution has higher solute concentration and lower water concentration. Water moves down its concentration gradient.

S5. How does the sodium-potassium pump maintain resting nerve cells?

It uses ATP to actively move Na+ out of the cell and K+ into the cell, maintaining higher K+ and lower Na+ inside the cell.

S6. Differentiate between phagocytosis and pinocytosis.

Phagocytosis takes in solid material, while pinocytosis takes in liquids in the form of droplets.

S7. How does receptor-mediated endocytosis ensure specificity?

Specific receptor proteins on the plasma membrane recognize and bind particular molecules, such as LDL cholesterol, before vesicle formation.

S8. Why is exocytosis important for secretory cells?

It allows bulky products such as proteins and hormones to be packaged in vesicles and released outside the cell by fusion with the plasma membrane.

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TOPIC 3.8 — STEM CELLS

Textbook topic 3.8 · printed pages 87–88

Q.17 Describe the categories of stem cells and their uses in medicine.

Board long question · 8 marks

Categories based on differentiation potential

Type	Potential	Example
Totipotent	Can differentiate into all possible cell types	Zygote and cells from first few divisions
Pluripotent	Can turn into almost any cell	Cells from early embryo
Multipotent	Can differentiate into a closely related family of cells	Hematopoietic stem cells
Oligopotent	Can differentiate into a few different cell types	Adult lymphoid or myeloid stem cells
Unipotent	Can produce only one cell type but can self-renew	Adult muscle stem cells

Categories based on origin

- **Embryonic stem cells (ESCs):** Derived from the inner cell mass of blastocysts; pluripotent; ethical concerns include use of embryos, teratoma risk and immune rejection.
- **Adult stem cells (ASCs):** Found in bone marrow, fat and blood; multipotent; less ethical controversy and lower immune rejection if from the patient, but limited potential.
- **Induced pluripotent stem cells (iPSCs):** Generated by reprogramming adult somatic cells to a pluripotent state; avoid embryo destruction and can be patient-specific.

Uses

1. **Regenerative medicine:** Repair or replace damaged tissues/organs, e.g., spinal cord injuries, type 1 diabetes, Parkinson's disease and heart disease.
2. **Drug testing and development:** Create models of human diseases for accurate drug testing.
3. **Personalized medicine:** Derived from a patient's own cells, reducing immune rejection.

Important Conceptual Short Questions — Topic 3.8

S1. Why are totipotent stem cells more versatile than pluripotent stem cells?

Totipotent cells can differentiate into all possible cell types, including extra-embryonic tissues, while pluripotent cells can form almost any cell type but not all.

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S2. How do iPSCs avoid the ethical issues of embryonic stem cells?

They are generated by reprogramming adult somatic cells, so they do not require destruction of embryos.

S3. Why are adult stem cells less likely to cause immune rejection?

When derived from the patient's own tissues, they are genetically matched to the patient, reducing the risk of immune rejection.

S4. How are stem cells used in regenerative medicine?

They can repair or replace damaged tissues and organs, offering treatment for conditions such as spinal cord injuries, diabetes, Parkinson's disease and heart disease.

S5. Differentiate between multipotent and unipotent stem cells.

Multipotent stem cells can differentiate into a closely related family of cells, while unipotent stem cells can produce only one cell type but can still self-renew.

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